Image dataset requirements

Format of images used as input

MicroICS requires 3D biofilm images (z-stacks) as input, with each image comprising multiple layers representing different depths within the biofilm. Images must be saved in TIFF format. Each input file must correspond to a single imaging position and include all z-layers for that position as one multi-page TIFF stack. The method for creating this format will depend on your microscope and acquisition software. Two common scenarios are described below:

Zeiss microscopes (using “positions” and “hardware autofocus” as described in Janež, Škrlj, Osojnik et al., 2026): when imaging multiple wells and positions in a microtiter plate, Zeiss microscopes may save all images in a single file, similar to a hyperstack. These must be split into individual TIFF files before analysis – one file per position, containing all z-layers. This can be achieved using the open-source software ImageJ/Fiji (https://imagej.net/software/fiji/) with the provided ImageJ script for splitting and converting LSM files (see Figure 1) available here: 10.5281/zenodo.20377151.

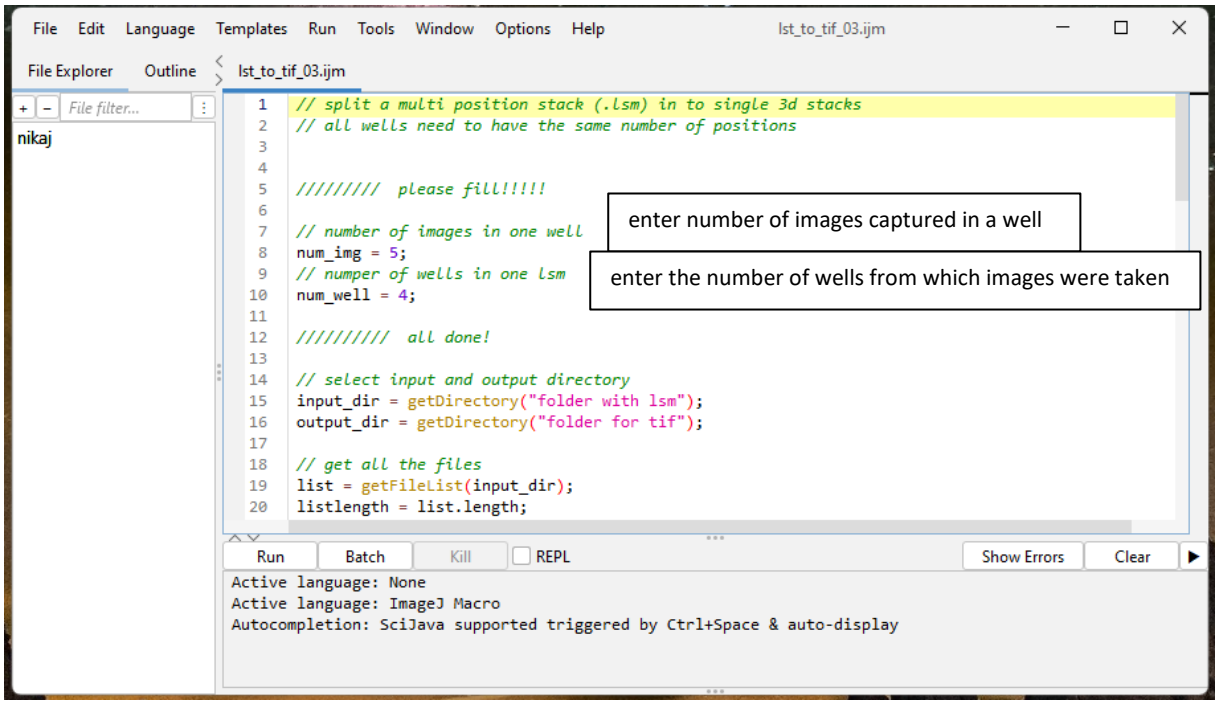


Figure 1. A screenshot of the script “lsm_to_tif,” available at 10.5281/zenodo.20377151, is shown. The script enables the conversion of a lsm file with several 3D stacks taken at different positions in microtiter plate into a single 3D stack per position (one-position--one-set of images – one 3D stack) in tif format. The script is loaded by dragging it onto the ImageJ; once it opens, the number of images per well and the number of wells in which images are captured must be adjusted. This script also relies on the image naming shown below. If you do not follow these, the script will not work.

Other microscopes: ideally, each imaged position is saved as an individual TIFF stack and named as described below. If the images are saved in a different proprietary format, ImageJ/Fiji supports a wide range of formats through the Bio-Formats plugin and can be used to export images as multi-page TIFF stacks before analysis.

Image file naming convention

The execution of the MicroICS pipeline depends entirely on the information contained in the image names, which should follow the example below.

Scheme for image naming:
<date>_s_<species>_st_<strain>_p_<well_coordinates>_pos<well_position>_tm_<growth_time>_ch_<channel>_z_<layer_distance>

Example of an image name: 07112023_s_Lm_st_L1823_p_F02_pos005_tm_24_ch_Syto9_z_21

Explanation of the name:

- 07112023: Date currently included in DayMonthYear to make it easier to track results
- s_Lm: s_ indicates the species, Lm means *Listeria monocytogenes*
- st_L1823: st_ denotes the strain, L1823 is the name of the strain (for the task strain identity). **This is the part of the filename the pipeline reads as the label.**
- p_F02: p_ denotes the well coordinates on the 96-well plate, F02 is the actual position on the 96-well plate
- pos005: pos denotes the measured position within the individual well, e.g., position 5 in well F02
- tm_24: tm_ stands for the measurement time, 24 means that the images were taken after 24 hours of biofilm growth
- ch_Syto9: ch_ denotes the channel, Syto9 is the name of the dye and filter used
- z_21: z_ denotes the number of layers in the 3D image, which also reflects the distance between two consecutive layers.

The most important parts of the naming structure are: Date, _s_, _st_, _p_, _pos. The names are parsed on the back-end according to these patterns. Any deviation from the exact patterns listed will result in failed or incorrect processing of the data.
The renaming process can be carried out semi-automatically by the software, e.g. https://www.advancedrenamer.com/, or by user-defined scripts. Examples and the user manual are available here https://www.advancedrenamer.com/user_guide/v3/complete_guide together with graphical representations.

Unit of replication

The unit of replication means you need to define what counts as an independent measurement in your particular case: is it a date, a well, or a position? The model uses this information to test its performance during training, when performance is being evaluated: When splitting the data, the pipeline keeps all images from the same unit together, so that images from one biofilm are never divided between training and testing, which would otherwise inflate the apparent accuracy. To carry out the evaluation properly in terms of the unit of replication, enough replicates per unit must be available, and once you load the dataset, the system reports whether the evaluation can be done based on the proposed unit of replication. This is feedback on computational feasibility, but not on whether the choice is biologically correct.

Number of images per class

Ensure that your dataset contains roughly equal numbers of images per class. This affects model training and evaluation.